# Shayan Sadigh

**Location**: Santa Clara, CA
**Email**: ssadigh.contact@gmail.com
**Github**: https://github.com/shayanpersonal (200+ stars)
**Work Status**: US Citizen

## Summary

- Founding Engineer, Agentic AI @Aurascape.ai | Previously @Amazon SageMaker
- Placed top-10 in $1.5 million AI competition sponsored by Department of Homeland Security. (Slides)
- Authored "checkered subsampling" deep learning research paper, published PyTorch code. (arXiv)

## Work Experience

**Aurascape.ai**
Founding Engineer, Agentic AI (Feb. 2024 - Current)

- Presented demos and drafted patents which helped raise $12.8 million funding.
- Developed agentic web browser implementing set-of-marks prompting (https://arxiv.org/abs/2310.11441), an agentic web traffic analyzer, and multimodal content classifiers.
- Leveraged LLM APIs such as OpenAI, Anthropic, Gemini, and in-house vLLM deployment. Built and deployed Docker containers to Google Cloud, AWS, Azure. Integrated with MCP protocol.

**Santa Barbara City College**
Adjunct Faculty (Aug. 2023 - Dec. 2023)

- Taught in-person 16-week course on Python, machine learning, and deep learning. Labs included a "small language model" neural network trained with PyTorch on tiny-shakespeare dataset.
- Incorporated MoeBot, my previously built agentic AI, as an AI teaching assistant.
- Quote from department chair: "The student evaluation ratings are the best I've ever seen out of the 12 years of evaluations packets I've signed off on."

**Moebot.ai**
Entrepreneurship (2021-2023)

- Built MoeBot, a subscription-based Discord AI agent powered by a swappable LLM backend. Incorporates RAG, agent memory, tool calling, Stripe payments. Deployed to Google Cloud.
- Built various other projects such as MIDI recording tools and VST audio plugins in Rust, Python, C.

**Amazon Web Services**
Deep Learning Software Engineer (Mar. 2020 - Dec. 2020)

- Worked with researchers to build AI at Amazon. Helped integrate Apache TVM deep learning compiler into Amazon SageMaker's PyTorch and XGBoost forks in Python and C++.

**Palo Alto Networks**
Senior Staff Software Engineer (Oct. 2019 - Mar. 2020)
Staff Software Engineer (Sep. 2018 - Sep. 2019)

- Awards: Project of the Quarter - Machine Learning Antivirus
- Proposed and implemented new stack-based pattern matching algorithm in C which improved threat coverage in PAN-OS firewall, resulting in higher scores on 3rd-party benchmarks.

- Solved many long-standing tickets from customers, data science, product managers, and threat research. Quote from VP of Engineering: "One-man army."
- Made major contributions to industry-first machine learning features - integrated inline machine learning classifiers into resource-constrained firewall in C.
- Prepared and gave presentation explaining the pattern matching system in the threat detection codebase. The presentation recording is now used to train new hires and cross-functional teams working with threat detection code.
- Brought up and helped train two junior engineers.

## Projects / Publications

- stacked-autoencoder-pytorch (100+ stars) - a popular PyTorch autoencoder influenced by Hinton's 2006 "deep belief nets" paper.
- Checkered CNN (30+ stars) - a deep learning technique I proposed, checkered subsampling.
- Passenger Screening Model (20+ stars) - my top-10 solution to a $1.5 million Kaggle challenge.
- GauchoMap Chrome Extension - 200+ users when I was maintaining it (2016).

## Education

- 5-year accelerated B.S. + M.S. program @ University of California, Santa Barbara
- Master of Computer Science (2017 - 2018)
- Bachelor of Computer Science (2013 - 2017)

## Miscellaneous

- 2nd-degree black belt in Taekwondo. Engaging in Brazilian Jiu-Jitsu and Muay Thai.
- Playing and writing piano music since early childhood.
- Expert Dance Dance Revolution player.
- Ranked #1 in online arena shooter.
- Languages: English (native), Farsi (native, but very rusty)